Hardik Thapar

Chandigarh, India | +91 98724-05709 | hardikthapar1@gmail.com
linkedin.com/in/hardik-thapar | github.com/hardik-thapar

Software Engineer Intern | Backend & Distributed Systems | Applied ML

EDUCATION

Chitkara University, Punjab, India Expected Aug 2028
B.E. Computer Science (AI & ML), CGPA 9.3/10
Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

TECHNICAL SKILLS

Languages: Go, Java (250+ LeetCode problems), Python, SQL, C++
Backend: REST APIs, FastAPI, Gin, PostgreSQL, SQLAlchemy, concurrency, idempotency, fault tolerance
Machine Learning (ML): scikit-learn, CatBoost, Librosa, time-series forecasting, feature engineering
Tools & Cloud: Docker, Docker Compose, Git, Linux, testcontainers-go, integration testing

PROJECTS

Dispatch: Fault-Tolerant Job Queue github.com/hardik-thapar/dispatch
Go, PostgreSQL, Gin, Docker

- Built a Postgres-backed job queue in Go with a REST API: FOR UPDATE SKIP LOCKED claims, leases, heartbeats, attempt-number fencing, exponential backoff with jitter, dead-lettering and replay, idempotent submit.
- About 4,600 jobs/s peak with no-op jobs (16 worker processes, Apple-silicon laptop); p99 queue wait of 18 ms or less at 500 to 2,000 jobs/s arrival rate.
- Wrote a chaos harness that runs kill -9 on workers under load. Across 6,000 jobs and 60 crashes: 0 lost jobs; the idempotent handler re-ran 117 jobs with 0 duplicate effects (non-idempotent control: 120); median recovery 2.9 s at a 3 s lease.
- Benchmarked design choices: SKIP LOCKED gave 0 double claims vs 102,921 for select-then-update; a partial index cut a claim query from 60.7 ms to 0.006 ms on 1,000,000 rows; jitter cut peak retry rate from 10,000 to 1,294 per second (simulated).
- Tested with testcontainers-go and -race; every number reproduces with make bench and make chaos.

VocalWell: Voice Analysis Backend github.com/hardik-thapar/vocalwell_backend
Python, FastAPI, PostgreSQL, Docker Compose

- Built a containerized FastAPI REST backend (upload, list, get by ID, PDF report, health check) that serves a pre-trained TensorFlow CNN for three-class voice classification.
- Extracted acoustic features with Librosa, generated multi-page PDF reports with ReportLab, stored results in PostgreSQL (SQLAlchemy, raw SQL), and wrote an integration test script.

DengCast: Dengue Forecasting Pipeline github.com/hardik-thapar/dengcast-dengue-forecasting
Python, CatBoost, scikit-learn, pandas

- Built a weekly case forecasting pipeline on DrivenData DengAI data with lag features (1 to 4 weeks), rolling statistics and a log-transformed target.
- Used chronological TimeSeriesSplit to avoid leakage. Cross-validated MAE: 11.87 (San Juan) and 4.74 (Iquitos), against baselines of about 30 and 31.

EXPERIENCE

National Kaohsiung University of Science and Technology (NKUST), Taiwan (remote) Mar 2026 – May 2026
Research Intern, Software & Data Systems

- Reviewed literature on EEG-based Alzheimer's detection and leave-one-subject-out evaluation, under a professor's supervision.
- Built Python preprocessing and feature-extraction pipelines for 19-channel resting-state EEG (OpenNeuro ds004504, 65 subjects) and ran experiments through them.
- Took part in designing a hybrid EEG model with the professor and team; PhD students led the deep-network implementation. The lab's results show epoch-level evaluation inflates accuracy by about 9.4 percentage points.

HACKATHONS & RECOGNITION

- **National Finalist:** AMD AI Sprint, IIT Bombay (top 150 of 4,000; AI scheduling agent on AMD MI300); GSMA Network-as-Code Hackathon, IMC 2025 (top 50 of 2,000).
- **1st place:** Swasthathon 2025, PCI India (436 teams, Rs. 1,00,000); Hackaccino 3.0, EduChain track (300+ teams); HackEMinds 2025 (international). **Also:** Top 5, CodePi 2025 (600+ entries); 2nd, Code With DCG '25, DTU; winner, GDG Hacks; 1st at 3 Chitkara University events.
- **Programs:** AWS AI & ML Scholars 2025 (completed Challenge Phase); McKinsey Forward Program 2025–26; Harvard Aspire Leaders Program 2025; CS50x; Microsoft Azure AI Fundamentals.